

Technical Datasheet IO Modules with Modbus RTU Protocol with RS485 Interface

The I/O modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (this feature however also depends on the signal strength of the Modbus Master Device).

The RS485 Digital Sensor module is sturdy, low power usage and easy to use.

2 PT100/1000-2 Thermocouple Module:-



Our Multi-Channel Temperature Measurement System is a versatile and high-precision solution designed for diverse temperature sensing needs. The system integrates Modbus RS485 communication for robust and reliable data transmission. With dual channels for RTD (PT100/PT1000) measurement and dual channels for thermocouple measurement, this device is ideal for industrial, commercial, and scientific applications requiring precise temperature monitoring and remote data access.

Dual RTD Channels:-

1. Features two channels for RTD measurements, capable of reading both PT100 and PT1000 sensors.
2. Utilizes advanced analog-to-digital conversion technology for accurate and reliable temperature readings.

Dual Thermocouple Channels:-

1. Includes two channels for thermocouple measurements, compatible with various thermocouple types (**K, J, N, R, S, T, E, B**).
2. Ensures precise temperature monitoring across a wide range of applications.
3. The I/O modules are housed in a DIN rail mountable casing with exposed connectors and LED indicators for easy status monitoring. The DIP switches for setting the Slave ID and Baud rate are conveniently located inside the enclosure.
4. The modules feature resettable fuses to protect against reverse polarity connections on both the power and communication ports, ensuring safety and reliability.

Specifications

General:-

Connectors

2 Pin 5.08 mm pitch pluggable screw terminals.

Dimensions

70 mm L x 110 mm B x 50 mm H

Power

Input Power – 24 V AC/DC

Typical – 12V DC @ 80mA

Operating Temperature

0 – 60° C (32 ~ 140°F)

Storage Temperature

-20 - 70° C (-4 ~ 158°F)

Storage Humidity

5 ~ 95 % RH, non – Condensing

Certifications



AI Inputs:-

Channels

4

Input Signal

2–PT100/PT1000–(jumper selectable)-Two wire/Three wire
2–Thermocouple – (K, J, N, R, S, T, E, B).

Accuracy

Total Accuracy over All Operating Conditions: 0.5NC
(0.05% of Full Scale) max.

Input Resolution

15-Bit ADC Resolution for PT's And 19-Bit ADC
Resolution for Thermocouple.

Isolation

Optically Isolated.

Additional Features:-

Communication port isolated

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air

EFT IEC 61000-4-4, 50A (5/50ms)

750V isolation.

CRC Error check.

No configuration needed on the IO board

Configuration Settings:-

Communication Speed 9600 – 19200 Kbps (DIP SW Selectable)

Data Bits 8

Parity None

Stop bit 1

CRC Yes

Slave ID 1-15 (DIP SW Selectable)

Function Code 0x03 Read Holding Registers

AI Register Address 0,1,2,3

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
0	RTD1	40001	0X03	RS485	16 Bit Unsigned int
1	RTD2	40002	0X03	RS485	16 Bit Unsigned int
2	THR1	40003	0X03	RS485	16 Bit Unsigned int
3	THR2	40004	0X03	RS485	16 Bit Unsigned int

Recommended Cable Electrical Characteristics:-

22 AWG Cable

Tinned Copper

Nominal Conductor DCR

Nominal Capacitance

High Frequency Non-Insertion Loss

Shielded and twisted pair should be used

Recommended

14.7 ohm / 1000 ft

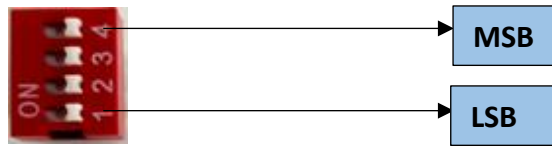
11 pf / feet (conductor to conductor)

0.5db / 100ft

Note:-

For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

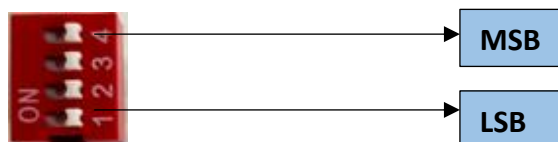
BAUD RATE DESCRIPTION:-



- For Baud rate Selection, DIP SW is used as per the diagram.
- Pulling up the switch will make Baud rate active.
- If no selection is made 9600 will be default Baud rate.
- When you change the Baud rate in the Module power 'ON' condition, please press the reset button to get Change to affect.

Baud Rate	DIP SWITCH			
	1	2	3	4
9600	OFF	OFF	OFF	OFF
19200	OFF	OFF	OFF	ON
38400	OFF	OFF	ON	OFF
57600	OFF	ON	OFF	OFF
115200	ON	OFF	OFF	OFF

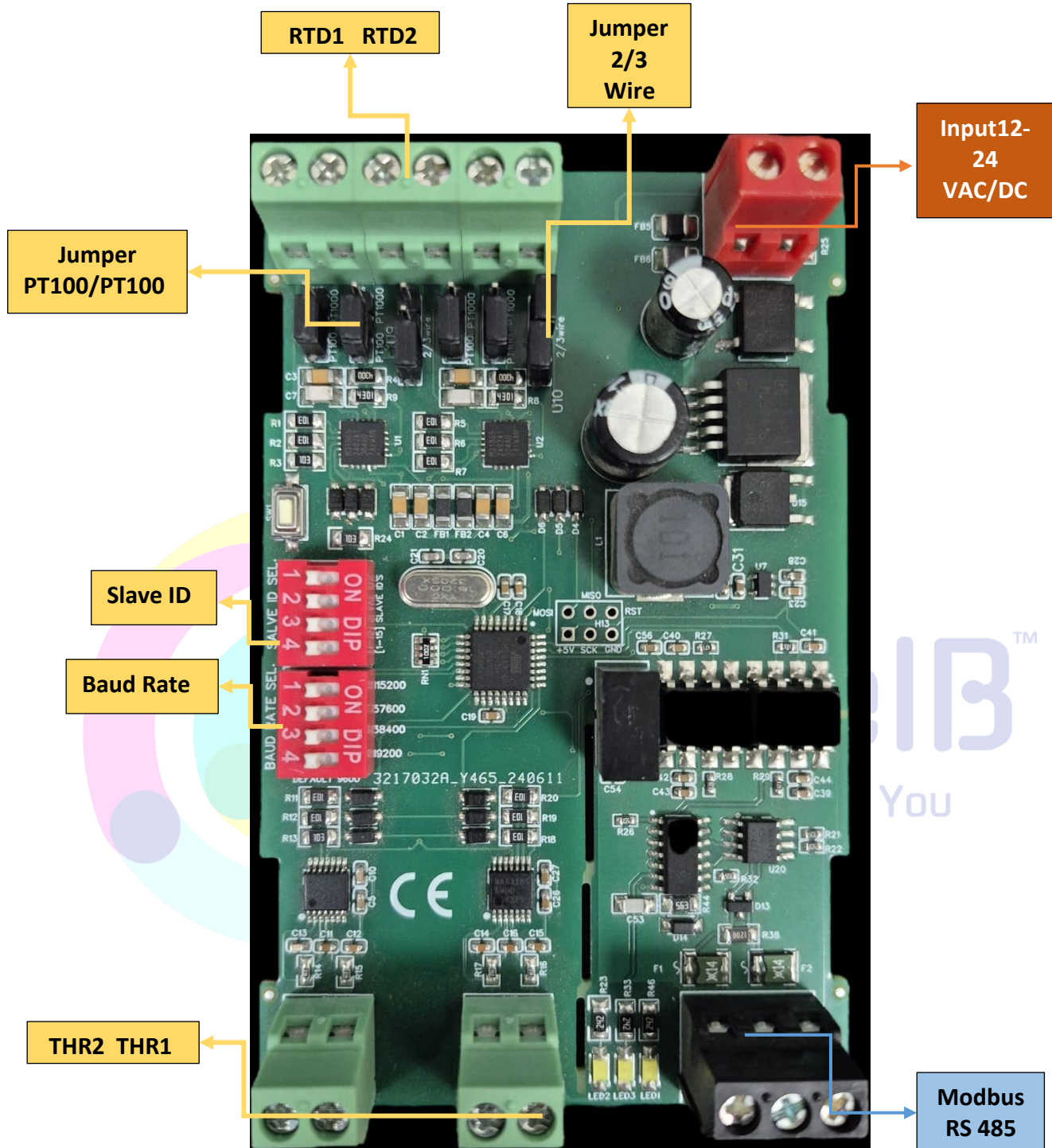
SLAVE ID DESCRIPTION:-



- For Slave ID Selection SW is used to Set The SLAVE ID .
- For Slave ID DIP Switch **LSB is "1"** follow through **"4" is MSB**.
- Slave ID Confirmed through below Device ID table .
- IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So "1 0 0 0" will be selected as Slave ID 1.

Slave ID	DIP SWITCH				OUTPUT (Binary)	OUTPUT (Decimal)
	1	2	3	4		
0	OFF(0)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
1	ON(1)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
2	OFF(0)	ON(1)	OFF(0)	OFF(0)	0 0 1 0	2
3	ON(1)	ON(1)	OFF(0)	OFF(0)	0 0 1 1	3
4	OFF(0)	OFF(0)	ON(1)	OFF(0)	0 1 0 0	4
5	ON(1)	OFF(0)	ON(1)	OFF(0)	0 1 0 1	5
6	OFF(0)	ON(1)	ON(1)	OFF(0)	0 1 1 0	6
7	ON(1)	ON(1)	ON(1)	OFF(0)	0 1 1 1	7
8	OFF(0)	OFF(0)	OFF(0)	ON(1)	1 0 0 0	8
9	ON(1)	OFF(0)	OFF(0)	ON(1)	1 0 0 1	9
10	OFF(0)	ON(1)	OFF(0)	ON(1)	1 0 1 0	10
11	ON(1)	ON(1)	OFF(0)	ON(1)	1 0 1 1	11
12	OFF(0)	OFF(0)	ON(1)	ON(1)	1 1 0 0	12
13	ON(1)	OFF(0)	ON(1)	ON(1)	1 1 0 1	13
14	OFF(0)	ON(1)	ON(1)	ON(1)	1 1 1 0	14
15	ON(1)	ON(1)	ON(1)	ON(1)	1 1 1 1	15

NOTE:-By default, the Jumper setting on the board is PT100.



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